

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

MYLAN PHARMACEUTICALS INC., TEVA PHARMACEUTICALS
USA, INC., and INVAGEN PHARMACEUTICALS, INC.

Petitioners,

v.

BAYER PHARMA AKTIENGESELLSCHAFT,

Patent Owner.

IPR2022-00517¹
Patent 10,828,310 B2

Before TINA E. HULSE, ROBERT A. POLLOCK, and RYAN H. FLAX,
Administrative Patent Judges.

HULSE, *Administrative Patent Judge.*

JUDGMENT
Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

¹ IPR2022-01513 and IPR2022-01515 have been joined with this proceeding.

I. INTRODUCTION

Mylan Pharmaceuticals Inc. (“Mylan”) filed a Petition requesting *inter partes* review of claims 1–8 of U.S. Patent No. 10,828,310 B2 (Ex. 1001, “the ’310 Patent”), owned by Bayer Pharma Aktiengesellschaft (“Patent Owner”). Paper 1 (“Pet.”). Upon considering the Petition, Preliminary Response (Paper 11), Petitioner’s Reply to Patent Owner’s Preliminary Response (Paper 13), and Patent Owner’s Sur-Reply (Paper 14), we instituted an *inter partes* review of the challenged claims of the ’310 patent. Paper 15 (“Dec. Inst.” or “Institution Decision”).

Teva Pharmaceuticals USA, Inc. (“Teva”) and Invagen Pharmaceuticals, Inc. (“Invagen”) both filed substantively identical petitions to Mylan’s Petition along with Motions for Joinder to join this proceeding. IPR2022-01513, Papers 1, 3 (Teva Petition and Motion); IPR2022-01515, Papers 1, 3 (Invagen Petition and Motion). In light of the Stipulation Concerning Joinder reached by the parties (*see, e.g.*, IPR2022-01513, Paper 9), we granted both petitions and joined those proceedings with this proceeding. *See* Papers 32 (Invagen), 33 (Teva). For the sake of simplicity, we collectively refer to Mylan, Teva, and Invagen as “Petitioner,” unless stated otherwise.

After institution, Patent Owner filed a Response to the Petition (Paper 36, “PO Resp.”), Petitioner filed a Corrected Reply (Paper 52, “Pet. Reply”), and Patent Owner filed a Sur-reply (Paper 56, “PO Sur-reply”).

Petitioner also filed a Motion to Exclude Evidence (Paper 59), to which Patent Owner filed an Opposition (Paper 61), and Petitioner filed a Reply (Paper 62).

An oral hearing was held on May 10, 2023. A transcript of the hearing has been entered in the record. Paper 67 (“Tr.”).

We have authority under 35 U.S.C. § 6. We issue this Final Written Decision under 35 U.S.C. § 318(a) and 37 C.F.R. § 42.73. For the reasons that follow, we determine Petitioner has shown by a preponderance of the evidence that claims 1–8 of the ’310 patent are unpatentable.

A. Real Parties-in-Interest

Mylan identifies itself and also Mylan Inc. and Viatris Inc. as real parties-in-interest. Pet. 1–2.

Teva identifies itself as the real party-in-interest of IPR2022-01513. *See* IPR2022-01513, Paper 2, 1.

Invagen identifies itself and Cipla (ET Limited), Cipla Limited, and Cipla USA Inc. as the real parties-in-interest of IPR2022-01515. *See* IPR2022-01515, Paper 1, 2.

Patent Owner (a/k/a Bayer AG) identifies itself and also Bayer Pharma AG and Janssen Pharmaceuticals, Inc. as real parties-in-interest. Paper 5, 2.

B. Related Proceedings

As of the June 26, 2023, filing of its Seventh Updated Mandatory Notices, Patent Owner discloses that it has asserted the ’310 patent in the following proceedings, which Patent Owner identifies as matters that may affect, or may be affected by, a decision in this proceeding:

- *Bayer Pharma AG v. Lupin Limited*, No. 21-314-RGA (D. Del.) (consolidated) (“Lupin Action”)
- *Bayer Pharma AG v. Dr. Reddy’s Laboratories, Inc.*, No. 21-732-RGA (D. Del.), consolidated with the Lupin Action
- *Bayer Pharma AG v. Taro Pharmaceutical Industries Ltd.*, No. 21-1000-RGA (D. Del.), consolidated with the Lupin Action

- *Bayer Pharma AG v. Teva Pharmaceuticals USA, Inc.*, No. 21-1001-RGA (D. Del.), consolidated with the Lupin Action
- *Bayer Pharma AG v. Mylan Pharmaceuticals Inc.*, No. 21-99-TSK (N.D.W.V.), which was transferred to the U.S. District Court for the District of Delaware in connection with the multi-district litigation proceedings identified below and is now designated as No. 21-1742-RGA (D. Del.)
- *Bayer Pharma AG v. Mylan Pharmaceuticals Inc.*, No. 22-63-JPB (N.D.W.V.), which was transferred to the U.S. District Court for the District of Delaware in connection with the multi-district litigation proceedings identified below and is now designated as No. 22-1228-RGA (D. Del.)
- *In re: Xarelto (Rivaroxaban) ('310) Patent Litigation*, MDL No. 21-3017-RGA (D. Del.), multi-district litigation proceedings involving the actions identified above
- *Bayer Pharma AG v. Apotex Inc.*, No. 22-01596-RGA (D. Del.)
- *Bayer Intellectual Property GmbH v. Biocon Pharma Ltd.*, No. 23-00334-RGA (D. Del.)
- *Bayer Pharma AG v. Dr. Reddy's Laboratories, Inc.*, No. 23-00410-RGA (D. Del.)
- *Bayer Pharma AG v. Teva Pharmaceuticals USA, Inc.*, No. 23-00551-RGA (D. Del.)
- *Bayer Pharma AG v. Auson Pharmaceuticals Inc.*, No. 23-03020-EP (D.N.J.)
- *Bayer Pharma AG v. Macleods Pharmaceuticals Ltd.*, No. 23-00665-RGA (D. Del.)
- *Bayer Intellectual Property GmbH v. Indoco Remedies Limited*, No. 23-00690-UNA (D. Del.)

See Paper 69, 2–5; *see also* Paper 40, 2 (Petitioner’s updated mandatory notices listing related cases as of December 14, 2022).²

C. *The ’310 Patent*

The ’310 Patent relates to methods of treating or preventing cardiovascular events in patients with atherosclerotic vascular disease, coronary artery disease (“CAD”), and/or peripheral artery disease (“PAD”). Ex. 1001, 1:16–19. Both CAD and PAD are caused by atherosclerosis, which occurs when plaque builds up in arteries and narrows or blocks blood flow to various regions of the body. *Id.* at 1:65–67. According to the ’310 Patent, CAD is the most common type of heart disease and occurs when plaque builds up in the heart’s arteries. *Id.* at 1:49–54. If blood flow to the heart becomes reduced or blocked, angina (i.e., chest pain) or a myocardial infarction (i.e., heart attack) may occur. *Id.* at 1:54–55. PAD results from narrowing of the arteries to the limbs and head, causing symptoms such as pain in the legs or buttocks when exercising, or causing a stroke if there is blockage of vessels in the neck. *Id.* at 1:57–62.

Patients with CAD or PAD are at high risk for major cardiovascular events, such as myocardial infarction, stroke, and cardiovascular death. *Id.* at 2:1–3. Antiplatelet therapy, such as aspirin alone or in combination with newer antiplatelet agents such as clopidogrel, is often used in patients with CAD. *Id.* at 2:23–26. In PAD, the “mainstay of treatment” includes use of a

² We note Petitioner’s Updated Mandatory Notice of December 14, 2022, identified *Bayer Pharma AG v. USV Private Ltd.*, No. 22-05485-EP (D.N.J.) as a related case. Paper 40, 2. The *Private* case appears on Patent Owner’s Sixth Updated Mandatory Notice (Paper 68, 3), but not the Seventh (Paper 69). We assume the *Private* case settled or terminated between the filings of Patent Owner’s Sixth Mandatory Notice and its Seventh.

single antiplatelet agent daily to prevent major cardiovascular events. *Id.* at 2:15–17.

Rivaroxaban is a selective inhibitor of coagulation factor Xa and serves as an anticoagulant for treatment and prevention of thromboembolic disorders. *Id.* at 1:23–42. According to the '310 Patent, “[a]nticoagulant therapies until the present invention have not been shown to the satisfaction of governmental health authorities to be superior to antiplatelet therapy in PAD and have been rejected as having unacceptably high rates of major bleeding.” *Id.* at 2:3–7. As such, the '310 Patent states “there is still a need for new approaches to improve health outcomes in CAD and PAD patients.” *Id.* at 2:27–29.

The '310 Patent describes a study that evaluated the potential benefit of rivaroxaban in patients with cardiovascular disease called the “Anti-Xa Therapy to Lower Cardiovascular Events in Addition to Standard Therapy in Subjects with Acute Coronary Syndrome 2—Thrombolysis in Myocardial Infarction 51” (ATLAS ACS 2-TIMI 51) trial (i.e., “the ATLAS trial”). *Id.* at 3:12–17. The '310 Patent states the ATLAS trial administered 2.5 mg or 5 mg of rivaroxaban twice daily in addition to single or dual antiplatelet therapy in patients with recent acute coronary syndrome. *Id.* at 3:18–21. The dose of 2.5 mg rivaroxaban twice daily resulted in a lower rate of major adverse cardiovascular events and lower mortality than placebo. *Id.* at 3:21–24.

The '310 Patent states that the findings presented in the Specification were obtained from a phase III clinical trial “evaluating the efficacy and safety of rivaroxaban (Xarelto®) for the prevention of major adverse cardiac events including cardiovascular death, myocardial infarction and stroke in patients with CAD or PAD” (referred to as “the COMPASS trial”). *Id.*

at 3:27–31. The Specification states the results are also reported in the *New England Journal of Medicine* in an article entitled, “Rivaroxaban with or without aspirin in stable cardiovascular disease.” *Id.* at 3:32–34. According to the Specification, “[t]he invention concerns the discovery that combination therapy of rivaroxaban and aspirin administered to patients with stable atherosclerotic vascular disease, including coronary artery disease and/or peripheral artery disease, shows efficacy in reducing the risk of myocardial infarction, stroke, and/or cardiovascular death.” *Id.* at 3:47–52. The Specification also states that “this combination therapy does not result in unacceptable bleeding, such as an unacceptably high risk of fatal bleeding or bleeding in critical organs.” *Id.* at 3:52–55.

D. Illustrative Claim

Petitioner challenges claims 1–8 of the ’310 Patent, of which claims 1 and 5 are independent. Claim 1 is illustrative and is reproduced below:

1. A method of reducing the risk of myocardial infarction, stroke or cardiovascular death in a human patient with coronary artery disease and/or peripheral artery disease, comprising administering to the human patient rivaroxaban and aspirin in amounts that are clinically proven effective in reducing the risk of myocardial infarction, stroke or cardiovascular death in a human patient with coronary artery disease and/or peripheral arterial disease, wherein the rivaroxaban is administered in an amount of 2.5 mg twice daily and aspirin is administered in an amount of 75-100 mg daily.

Ex. 1001, 18:56–65. Independent claim 5 differs from claim 1 only in specifying that “a first product comprising rivaroxaban and aspirin” and “a second product comprising rivaroxaban” are administered. *Id.* at 19:5–17.

Claims 2, 3, and 4 depend from independent claim 1 and recite the administered aspirin at an amount of “100 mg daily,” “81 mg daily,” and

“75 mg daily,” respectively. *Id.* at 18:66–19:4. Claims 6, 7, and 8 depend from independent claim 5 and further require the “first product” of claim 5 (i.e., a combination of rivaroxaban and aspirin) to comprise “75 mg aspirin,” “81 mg aspirin,” and “100 mg aspirin,” respectively. *Id.* at 19:18–24.

E. The Asserted Grounds of Unpatentability

Petitioner asserts that claims 1–8 of the ’310 Patent are unpatentable on the following grounds:

Claim(s) Challenged	35 U.S.C. § ³	Reference(s)/Basis
1–4	102	EMA ⁴
1–8	103	EMA
1–2	102	Foley ⁵
1–2, 5, 8	103	Foley
3–4, 6–7	103	Foley and Plosker ⁶

Petitioner relies upon, *inter alia*, the Declarations of Dr. Randall Zusman, M.D. (Ex. 1002; Ex. 1037). Patent Owner relies upon, *inter alia*, the Declaration of Dr. Jeffrey Weitz, M.D. (Ex. 2060).

II. ANALYSIS

Petitioner bears the burden of proving unpatentability of the challenged claims, and that burden of persuasion never shifts to Patent Owner. *Dynamic Drinkware, LLC v. Nat’l Graphics, Inc.*, 800 F.3d 1375,

³ Because the claims at issue have an effective filing date after March 16, 2013, the effective date of the applicable provisions of the Leahy Smith America Invents Act, Pub. L. No. 112–29, 125 Stat. 284 (2011) (“AIA”), we apply the AIA versions of 35 U.S.C. §§ 102 and 103 in this Decision. Our decision, however, is not impacted by which version of the statute applies.

⁴ European Medicines Agency, *Assessment Report, Xarelto*, EMA/CHMP/794349/2012 (2013). Ex. 1005 (“EMA”).

⁵ Foley et al., *Antithrombotic Therapy in Peripheral Artery Disease*, 21 VASCULAR MED. 156–69 (2016). Ex. 1006 (“Foley”).

⁶ Plosker, *Rivaroxaban: A Review of Its Use in Acute Coronary Syndromes*, 74 DRUGS 451–64 (2014). Ex 1007 (“Plosker”).

1378 (Fed. Cir. 2015). To prevail, Petitioner must prove unpatentability by a preponderance of the evidence.⁷ See 35 U.S.C. § 316(e); 37 C.F.R. § 42.1(d) (2019).

A. Person of Ordinary Skill in the Art

In determining the level of ordinary skill in the art, we consider the type of problems encountered in the art, the prior art solutions to those problems, the rapidity with which innovations are made, the sophistication of the technology, and the educational level of active workers in the field. *Custom Accessories, Inc. v. Jeffrey-Allan Indus., Inc.*, 807 F.2d 955, 962 (Fed. Cir. 1986).

Petitioner contends that a person of ordinary skill in the art (“POSA”) would have would have had “the equivalent of an M.D. degree, knowledge, and experience regarding the diagnosis and treatment of patients with CAD and PAD, including administering antiplatelet and anticoagulant agents to such patients.” Pet. 10 (citing Ex. 1002 ¶¶ 50–52, 89–95). Moreover, Petitioner contends that the POSA “could work in consultation with other specialists, including, *e.g.*, a clinical pharmacologist, hematologist, vascular surgeon, interventionalist, medicinal chemist, pharmaceutical chemist, statistician, or a pharmacist depending on each patient’s needs.” *Id.*

Patent Owner disagrees with Petitioner’s definition and contends that a POSA would have been a “clinician with experience in the design, conduct, analysis, and interpretation of results from clinical trials involving antithrombotic agents (anticoagulants and antiplatelet agents), as well as experience with the treatment of patients with anticoagulants and antiplatelet

⁷ Although we find certain of Patent Owner’s arguments not persuasive, this should never be taken as an indication that we have assigned Patent Owner the burden of proof on patentability.

agents alone or in combination, including experience with the treatment of patients with coronary artery disease and/or peripheral artery disease.”

PO Resp. 22 (citing Ex. 2060 ¶¶ 24–26).

In response, Petitioner asserts that its definition encompasses Patent Owner’s definition, as the experience of an M.D. includes the ability to interpret the results of clinical trials to evaluate therapies. Pet. Reply 2–3 (citing Ex. 1037 ¶ 7).

We find Petitioner has the better position. We agree with Petitioner and Dr. Zusman that an M.D. with experience treating patients who worked in consultation with a team of specialists would include the qualifications recited by Patent Owner. Ex. 1037 ¶¶ 7–11. We, therefore, adopt Petitioner’s definition of a POSA, which indicates a high level of skill in the relevant art and is consistent with the prior art’s demonstration of the level of ordinary skill in the art at the time of the invention. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001) (explaining that specific findings regarding ordinary skill level are not required “where the prior art itself reflects an appropriate level and a need for testimony is not shown” (quoting *Litton Indus. Prods., Inc. v. Solid State Sys. Corp.*, 755 F.2d 158, 163 (Fed. Cir. 1985))).

Even if there were a distinction between the parties’ respective definitions of the level of ordinary skill in the art, Patent Owner’s expert, Dr. Weitz, notes that his opinion would not change regardless of which definition is applied. *See* Ex. 2060 ¶ 26. Moreover, we find both Dr. Zusman and Dr. Weitz qualified to opine from the perspective of a POSA for purposes of this Decision as both possess at least the level of ordinary skill in the art. *See* Ex. 1002 ¶¶ 1–11; Ex. 2060 ¶¶ 1–11; *see also Kyocera Senco Indus. Tools Inc. v. Int’l Trade Comm’n*, 22 F.4th 1369,

1376–77 (Fed. Cir. 2022) (“To offer expert testimony from the perspective of a skilled artisan in a patent case—like for claim construction, validity, or infringement—a witness must at least have ordinary skill in the art.”).

B. Claim Construction

In an *inter partes* review, the Board applies the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. § 282(b). 37 C.F.R. § 100(b) (2021). Under that standard, claim terms “are generally given their ordinary and customary meaning” as understood by a person of ordinary skill in the art at the time of the invention. *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc). “In determining the meaning of the disputed claim limitation, we look principally to the intrinsic evidence of record, examining the claim language itself, the written description, and the prosecution history, if in evidence.” *DePuy Spine, Inc. v. Medtronic Sofamor Danek, Inc.*, 469 F.3d 1005, 1014 (Fed. Cir. 2006) (citing *Phillips*, 415 F.3d at 1312–17). Extrinsic evidence is “less significant than the intrinsic record in determining ‘the legally operative meaning of claim language.’” *Phillips*, 415 F.3d at 1317.

1. “clinically proven effective”

Independent claims 1 and 5 each recite administering rivaroxaban and aspirin “in amounts that are clinically proven effective in reducing the risk of myocardial infarction, stroke or cardiovascular death in a human patient with coronary artery disease and/or peripheral arterial disease.” Ex. 1001, claims 1 and 5.

Petitioner asserts that because the claims recite administering rivaroxaban and aspirin “in amounts that are clinically proven effective,” those specific doses must inherently satisfy that functional element. Pet. 11

(citing Ex. 1002 ¶ 55). Petitioner also cites the '310 Patent Specification, which defines “effective” and “efficacy” as preferably referring to “a dosage determined by the US FDA as acceptable for administration to reduce the risk of major cardiovascular events (cardiovascular death, myocardial infarction, or stroke) in CAD and/or PAD patients.” Pet. 11–12 (quoting Ex. 1001, 10:57–63). Petitioner notes, however, that the FDA did not approve the claimed regimen until after the February 2018 priority date. *Id.* at 12 (citing Ex. 1016). Nevertheless, Petitioner argues that a POSA would have understood that approval by any major governmental health agency would require that the regimen was “clinically proven effective.” *Id.* (citing Ex. 1002 ¶ 59). Finally, Petitioner asserts that the claim phrase should be construed to encompass both individual and joint-efficacy. Pet. 12 (citing Ex. 1002 ¶ 56).

Patent Owner asserts that Petitioner’s construction is erroneous because a POSA would have understood that clinical proof of efficacy is only established in reference to a clinical study conducted in a defined group of patients. PO Resp. 33 (citing Ex. 2060 ¶¶ 96–97). According to Patent Owner, because the claims define the group of patients as those “with coronary artery disease and/or peripheral artery disease,” the group would include patients with CAD, and patients with PAD, and patients with both CAD and PAD. *Id.* (citing Ex. 2060 ¶¶ 98–100). Thus, Patent Owner asserts that clinical proof of efficacy must be with respect to the entire population of patients recited in the claim. *Id.* at 35 (citing Ex. 2060 ¶¶ 96–98). Moreover, Patent Owner asserts that a “POSA would understand that ‘clinically proven effective’ as used in the '310 patent requires data to support the notion that there is a statistically significant difference (*i.e.*, a *p*-value of less than 0.05) in the efficacy observed between the regimen at

issue (*e.g.*, rivaroxaban plus aspirin) compared to the standard of care (*e.g.*, aspirin). *Id.* at 37–38 (citing Ex. 2060 ¶ 94; Ex. 1001, Figs. 2A–4 (referencing *p*-values)).

In our Institution Decision, we preliminarily construed the term “clinically proven effective” to mean “the amounts of aspirin and rivaroxaban in combination that have clinically been shown to reduce the risk of myocardial infarction, stroke, or cardiovascular death in patients with CAD and/or PAD.” Dec. Inst. 14–15. We further held that the term does not require FDA approval, as the Specification states that the terms “effective” and “efficacy” only “[p]referably” refer to a dosage that has been approved by the FDA. *Id.* at 15; *see also* Ex. 1001, 10:57–59.⁸ Moreover, we held that, when reading the claims and the Specification as a whole, the term “clinically proven effective” refers to the amounts of aspirin and rivaroxaban in combination and not individually. *Id.* (citing Ex. 1001, claims 1 and 5, Abstract).

Having now considered the full trial record, we revise our preliminary construction and determine that the term “clinically proven effective” is nonlimiting, as Petitioner asserts. *See* Pet. 11; Pet. Reply 8–9, 16. We begin our analysis with the language of the claims, which recite administering rivaroxaban and aspirin in amounts that are “clinically proven effective,” “wherein rivaroxaban is administered in an amount of 2.5 mg twice daily and aspirin is administered in an amount of 75–100 mg daily.” Ex. 1001, claims 1 and 5. We are persuaded that a POSA would have understood from a plain reading of the claims that the specific recited “amounts” of

⁸ We also noted in our Institution Decision that in the Delaware MDL, the parties stipulated that claim 1 “does not require FDA approval.” Dec. Inst. 15 n.6 (quoting Ex. 2019, 2).

rivaroxaban and aspirin are “clinically proven effective.” *See* Ex. 1002 ¶ 55. That is, the independent claims expressly recite that the rivaroxaban is administered in an amount of 2.5 mg twice daily, and that aspirin is administered in an amount of 75–100 mg daily. Ex. 1001, claims 1 and 5. Because the claims expressly recite the amounts of rivaroxaban and aspirin, further identifying these amounts as “clinically proven effective” is redundant and unnecessary, and does nothing to alter the method steps of administering the pharmaceutical, as otherwise recited by these claims. *See Bristol-Myers Squibb Co. v. Ben Venue Labs., Inc.*, 246 F.3d 1368, 1375 (Fed. Cir. 2001) (finding claim term “antineoplastically effective” not limiting where express dosage amounts are material claim limitations that do not change based on the intended result).

We next consider the claim language in the context of the ’310 Patent Specification. The Specification states that “[t]he methods and products of the invention concern dosages that are clinically proven safe and effective.” Ex. 1001, 10:9–10. The Specification defines the terms “effective” or “efficacy” as referring to “reducing the risk of myocardial infarction, stroke, or cardiovascular death in patients with CAD and/or PAD or other atherosclerotic vascular disease, such as by reducing the risk when compared to therapy with aspirin alone.” *Id.* at 10:50–57. Although the Specification does not expressly define “clinically proven effective,” the Specification repeatedly describes administering rivaroxaban and aspirin “in amounts that are clinically proven effective,” “wherein rivaroxaban is administered in an amount of 2.5 mg twice daily and aspirin is administered in an amount of 75–100 mg daily, or in an amount of 75 mg, 81 mg or 100 mg daily.” *See, e.g., id.* at 3:59–67, 4:4–11, 4:28–35.

The Specification also describes the COMPASS study and its results. *Id.* at 13:35–18:54, FIGS. 2A–4 (referencing hazard ratios and *p*-values). The Specification explains that, in COMPASS, “a total of 27,395 persons . . . were randomly assigned to rivaroxaban plus aspirin, rivaroxaban, or aspirin.” *Id.* at 15:25–29. The Specification described the study results and stated:

In conclusion, in patients with stable atherosclerotic vascular disease, we compared three antiplatelet regimens: rivaroxaban (2.5 mg twice daily) plus aspirin (100 mg once daily), rivaroxaban (5 mg twice daily), and aspirin (100 mg once daily). The risk of major adverse cardiovascular events was significantly lower with the combination of rivaroxaban plus aspirin than with aspirin alone, and the risk of major bleeding was significantly higher.

Id. at 18:41–48. Despite the higher risk of major bleeding, the Specification states, “[o]ur definition of net clinical benefit balanced the lower risk of cardiovascular death, stroke, or myocardial infarction against the most serious bleeding events and showed a significant benefit of combination therapy.” *Id.* at 18:5–8.

Thus, reading the Specification as a whole, the ’310 Patent confirms that 2.5 mg twice daily rivaroxaban plus 75–100 mg aspirin daily, or in an amount of 75 mg, 81 mg or 100 mg daily are “clinically proven effective” amounts of rivaroxaban and aspirin.⁹ *See id.* at 3:59–67, 4:4–11, 4:28–35, 13:35–18:54.

⁹ We note that the Specification does not include clinical data on the efficacy of 75 mg or 81 mg daily aspirin, which further calls into question Patent Owner’s proposed construction, which would require actual data for those amounts to be considered “clinically proven effective.” *See* PO Resp. at 37–38

In light of that teaching, we determine that the phrase “clinically proven effective” in the claims is nonlimiting. We find *Bristol-Myers Squibb Co. v. Ben Venue Labs., Inc.*, 246 F.3d 1368 (Fed. Cir. 2001), to be on point. There, the claim at issue recited “[a] method for reducing hematologic toxicity in a cancer patient undergoing [t]axol treatment comprising parenterally administering to said patient an *antineoplastically effective amount* of about 135–175 mg/m² taxol over a period of about three hours.” *Id.* at 1371 (emphasis added). The Federal Circuit held that the term “antineoplastically effective amount” was nonlimiting, because it is an “expression of intended result [that] essentially duplicates the dosage amounts recited in the claims that are also described in the specification as ‘antineoplastically effective.’” *Id.* at 1375. The court continued, stating, “[t]he express dosage amounts are material claim limitations; the statement of the intended result of administering those amounts does not change those amounts or otherwise limit the claim.” *Id.* Moreover, “the claimed process here is not directed to a new use; it is the same use, and it consists of the same steps as described by [the prior art.] Newly discovered results of known processes directed to the same purpose are not patentable because such results are inherent.” *Id.* at 1376.

Similarly, here, the ’310 Patent claims recite the specific amounts of rivaroxaban and aspirin that are “clinically proven effective” (i.e., 2.5 mg rivaroxaban twice daily and 75–100 mg aspirin once daily). Just as the patent specification in *Bristol-Myers* taught that the claimed dosage amount was “antineoplastically effective,” 246 F.3d at 1375, the ’310 Patent Specification teaches that the claimed dosage amounts of rivaroxaban and aspirin are “clinically proven effective.” Ex. 1001, 3:59–67, 4:4–11, 4:28–35, 13:35–18:54. As such, like the term “antineoplastically effective” in

Bristol-Myers, the term “clinically proven effective” “does not change those amounts or otherwise limit the claim.” *See Bristol-Myers*, 246 F.3d at 1375. We, therefore, agree with Petitioner that “being subsequently proven clinically effective in COMPASS is at most simply a latent property of the prior-art method; nothing in the claimed method changes—no steps are added, subtracted, or modified for the prior-art patient—as a result of this knowledge.” Pet. Reply 8–9.

Patent Owner attempts to distinguish the *Bristol-Myers* case by arguing that the case did not involve an allegation that the results of a clinical trial were “inherent,” and did not involve the claim term “clinically proven effective,” or a similar requirement. PO Resp. 53. Although the facts of *Bristol-Myers* may not be identical, Patent Owner misses the point. As explained above, the holding of *Bristol-Myers* indisputably applies here—neither the term “effective” nor “clinically proven effective” changes the amounts of rivaroxaban and aspirin expressly recited in the claims. *See Bristol-Myers*, 246 F.3d at 1375. Indeed, Patent Owner does not argue otherwise.

Instead, Patent Owner asserts that Petitioner’s construction impermissibly reads the term “clinically proven” out of the claim. PO Sur-reply 13 (citing *Agilent Techs. v. Affymetrix, Inc.*, 567 F.3d 1366, 1377–78 (Fed. Cir. 2009)). The *Agilent* case, however, is inapposite. There, the court construed “chamber” to mean “an enclosed cavity defined by the inner surfaces of the first and second substrates,” and held that the term “closed” further required that “there is no egress of fluid” from the chamber. *Id.* at 1378. In contrast, here, “clinically proven” adds nothing to the term “effective”—both are nonlimiting statements of intended results that

duplicate the dosage amounts recited in the claims. *See Bristol-Myers*, 246 F.3d at 1375.

Patent Owner also relies on the Examiner's analysis in the Supplemental Examination as proof that the term "clinically proven effective" is "an important requirement of the claims." PO Resp. 53–54. As background, Patent Owner requested supplemental examination of claims 1–8 of the '310 Patent on January 7, 2021. Ex. 2001, 1576. Each of the references submitted by Patent Owner were obtained from the website ClinicalTrials.gov and relate to clinical trial NCT01776424 (i.e., the COMPASS study). *Id.* at 1577. The Examiner noted that the results of the COMPASS study were first published on ClinicalTrials.gov on October 5, 2018, after the February 2, 2017, critical date of the '310 Patent. *Id.* at 1577–78. Thus, to the extent any of the references disclosed study results, the results were published after the critical date. *Id.* at 1578. As such, the Examiner found the request for supplemental examination did not raise a substantial new question of patentability because "[n]one of the items of information discloses information regarding the efficacy of the claimed treatment methods, i.e. results of the clinical trial, which was available prior to the critical date of February 2, 2017." *Id.* at 1578–79.

Although we referred to the Examiner's finding in our Institution Decision as consistent with our Institution Decision (Dec. Inst. 28), that Decision was preliminary and based on a limited record. *Id.* at 29 (citing *TriVascular, Inc. v. Samuels*, 812 F.3d 1056, 1068 (Fed. Cir. 2016) (noting that "there is a significant difference between a petitioner's burden to establish a 'reasonable likelihood of success' at institution, and actually proving invalidity by a preponderance of the evidence at trial")). Having now considered the complete record of the arguments and evidence

presented at trial, we find the Examiner's construction to be inconsistent with Federal Circuit precedent, which the Examiner did not consider in his analysis. *See* Ex. 2001, 1577–79.

As noted above, our construction of “clinically proven effective” adopts Petitioner's proposed interpretation that it is not a limitation beyond the expressly recited amounts of rivaroxaban and aspirin recited in the independent claims. *See* Pet. 11–12, 29, 47. We have considered Patent Owner's remaining arguments regarding the alleged requisite population of patients studied (PO Resp. 33–37), but we find those arguments moot in light of our determination that “clinically proven effective” is a nonlimiting term of the claims.

2. “*first product*”

Independent claim 5 recites once daily administration of a “first product” comprising 2.5 mg rivaroxaban and 75–100 mg aspirin and a “second product” comprising 2.5 mg rivaroxaban. Ex. 1001, claim 5. Petitioner notes that the term “first product” and “second product,” other than in the claims themselves, do not appear in the '310 Patent Specification, and “product” is not expressly defined. Pet. 13. Petitioner's expert, Dr. Zusman, opines that a POSA would have understood the “first product” to be combination therapy, which the Specification describes “may be administered using separate dosage forms for rivaroxaban, or using a combination dosage form containing both rivaroxaban and aspirin.” Pet. 13 (emphasis omitted); Ex. 1002 ¶ 67 (quoting Ex. 1001, 8:65–9:20). Moreover, Dr. Zusman notes that the '310 Patent Specification states that “combinations” include “components administered simultaneously or sequentially as long as they are employed for the prophylaxis and/or treatment of the same disease.” Ex. 1002 ¶ 67 (quoting Ex. 1001, 8:65–

9:20). Accordingly, Petitioner asserts that a person of ordinary skill in the art “would have understood simply administering one 2.5 mg rivaroxaban pill and one aspirin pill at approximately the same time could constitute claim 5’s ‘first product.’” Pet. 14.

Patent Owner disagrees with Petitioner’s proposed construction of “first product.” PO Resp. 39. Patent Owner asserts that although the Specification discusses “combination therapy” as potentially administering “separate dosage forms for rivaroxaban and aspirin,” the Specification also identifies “combination therapy” as “using a combination dosage form containing both rivaroxaban and aspirin.” *Id.* (citing Ex. 1001, 8:66–9:1). Patent Owner also notes that the claims refer to the “first product” as “comprising rivaroxaban and aspirin.” *Id.* (citing Ex. 1001, 19:13–14). Thus, argues Patent Owner, a POSA would have understood “that the combination dosage form is what is clearly referred to in claim 5.” *Id.* (citing Ex. 2060 ¶¶ 103–105, 145–146).

We are not persuaded by Patent Owner’s argument. The parties appear to agree that the “first product” of claim 5 refers to “combination therapy,” which the Specification expressly states encompasses administering “separate dosage forms for rivaroxaban and aspirin” or “a combination dosage form containing both rivaroxaban and aspirin” in the recited amounts.¹⁰ See Ex. 1001, 8:26–9:1. Notwithstanding this broad language of the Specification, Patent Owner limits the term “first product” by equating it with a “combination dosage form.” The Specification, however, refers to “products and methods of the present invention”

¹⁰ In the Delaware MDL, the parties stipulated that claim 1 “does not require that the amounts of rivaroxaban and aspirin be in a single/combined dosage form (e.g., a tablet containing both rivaroxaban and aspirin).” Ex. 2019, 2.

separately from the “combination dosage form containing both rivaroxaban and aspirin.” *Compare* Ex. 1001, 12:28–29 *with* 8:65–9:1. Certainly if the patentee intended the “first product” to be a “combination dosage form,” it could have used the latter term. Instead, the Specification treats the term “product” as broader than “combination dosage form.” Consistent with that, the Specification states that “[c]ombinations’ mean for the purposes of the invention not only dosage forms which comprise all the components (so-called fixed dose combinations), and combination packs or kits which comprise the components separate from one another in a package together.” *Id.* at 9:9–14. Thus, we interpret the term “product” to include these various “combinations” described in the Specification, and decline to limit the term to a single dosage form, as Patent Owner urges. *Id.* Moreover, if administered as separate dosage forms, we agree with Petitioner that the Specification supports administering the “first product” dosage forms “simultaneously or sequentially as long as they are employed for the prophylaxis and/or treatment of the same disease.” *See id.* at 9:17–19.

Accordingly, we determine the “first product” of claim 5 is not limited to a single dosage form containing rivaroxaban and aspirin, and determine that if administered separately, the dosage forms can be administered simultaneously or sequentially as long as they are employed for the prophylaxis and/or treatment of the same disease. *See id.* at 9:17–19.

3. *Remaining Claim Terms*

Patent Owner also offers constructions for “coronary artery disease,” and the preamble (PO Resp. 23–32, 39–40), but none of these claim phrases

requires construction for purposes of our Decision.¹¹ *See Wellman, Inc. v. Eastman Chem. Co.*, 642 F.3d 1355, 1361 (Fed. Cir. 2011) (“[C]laim terms need only be construed ‘to the extent necessary to resolve the controversy.’” (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999))).

We now turn to the patentability challenges presented by Petitioner.

C. Anticipation by Foley

Petitioner asserts that claims 1 and 2 of the ’310 Patent are anticipated by Foley. Pet. 52–58. Patent Owner opposes. PO Resp. 51–54. Having considered the evidence and arguments presented at trial, we determine that Petitioner has established by a preponderance of the evidence that claims 1 and 2 are anticipated by Foley.

1. Foley (Ex. 1006)

Foley is a journal article entitled “Antithrombotic therapy in peripheral artery disease,” which “reviews current data on antithrombotic therapy in PAD and discusses the implications of this data for current practice and future research.” Ex. 1006, 156. Foley indicates it was published in 2016, which makes it prior art to the challenged claims, and Patent Owner does not assert otherwise. *Id.*; *see generally* PO Resp.

Foley discloses a variety of treatment regimens for patients with PAD, including the use of aspirin for symptomatic and asymptomatic PAD. Foley

¹¹ Indeed, we note that although Patent Owner proposes an alternative construction for “first product” than we set forth in our Institution Decision (Dec. Inst. 15–17), Patent Owner does not rely on that proposed construction in any of its substantive arguments on the merits. *See generally* PO Resp. 40–60; PO Sur-reply 15–25.

states “[d]espite the advent of newer agents, aspirin remains the most widely used antiplatelet drug in the world.” Ex. 1006, 157.

Foley also describes anticoagulant therapy in PAD. Specifically, Foley describes the Warfarin Antiplatelet Vascular Evaluation (WAVE) study. *Id.* at 164. In this randomized controlled trial, the study compared combination therapy with warfarin and an antiplatelet agent to an antiplatelet agent alone in patients with PAD. *Id.* The WAVE study found there was no significant difference between the treatment groups in the occurrence of heart attack, stroke or death from cardiovascular causes. *Id.* According to Foley, the WAVE findings “suggest that combination therapy with warfarin and an antiplatelet agent yields no significant benefit in patients with PAD when compared to antiplatelet therapy alone, and is associated with an increased risk of moderate and severe bleeding.” *Id.*

Finally, Foley discloses the COMPASS trial and states that it studies the effects of rivaroxaban in patients with CAD or PAD. *Id.* at 167. Foley states the COMPASS patients are randomized into one of three arms: (1) 2.5 mg rivaroxaban twice daily and aspirin (100 mg once daily); (2) 5 mg rivaroxaban twice daily and aspirin (same dosage); and (3) placebo twice daily and aspirin (same dosage). *Id.* Foley notes that the COMPASS trial is ongoing and estimated to be completed in February 2018. *Id.*

2. *Analysis*

“A claim is anticipated if each and every element as set forth in the claim is found, either expressly or inherently, in a single prior art reference.” *Arbutus Biopharma Corp. v. ModernaTX, Inc.*, 65 F. 4th 656, 662 (Fed. Cir. 2023). The Federal Circuit explains that “[a]nticipation by inherent disclosure is appropriate only when the reference discloses prior art that must *necessarily* include the unstated limitation.” *Monsanto Tech. LLC v.*

I.E. DuPont de Nemours & Co., 878 F.3d 1336, 1343 (Fed. Cir. 2018).

Inherent anticipation requires “merely that the disclosure of the prior art is sufficient to show that the natural result flowing from the operation as taught in the prior art would result in the claimed product.” *SmithKline Beecham Corp. v. Apotex Corp.*, 403 F.3d 1331, 1343–44 (Fed. Cir. 2005) (internal quotes omitted).

Petitioner asserts that Foley discloses each limitation of claims 1 and 2. Pet. 52–57. We agree. We find that Foley’s disclosure of the COMPASS clinical trial (i.e., Rivaroxaban for the Prevention of Major Cardiovascular Events in Coronary or Peripheral Artery Disease) and its protocol of administering to patients with CAD or PAD 2.5 mg rivaroxaban twice daily and 100 mg aspirin once daily discloses each limitation of claims 1 and 2. *See* Ex. 1006, 166–67.

That is, we are persuaded that a POSA would have understood that COMPASS’s study of preventing major cardiovascular events such as myocardial infarction, stroke, or cardiovascular death in coronary or peripheral artery disease constitutes a method of reducing the risk of those events in human patients with CAD and/or PAD, as required by the preamble.¹² Ex. 1006, 166–67; Ex. 1002 ¶ 250–254. We also find that Foley’s disclosure of administering 2.5 mg of rivaroxaban twice daily and 100 mg aspirin once daily discloses the step of “administering to the human patient rivaroxaban and aspirin in amounts that are clinically proven effective in reducing the risk of myocardial infarction, stroke, or cardiovascular death in a human patient with [CAD and/or PAD], wherein

¹² We need not expressly determine whether the preamble is limiting because Foley discloses the preamble regardless.

the rivaroxaban is administered in an amount of 2.5 mg twice daily and aspirin is administered in an amount of 75–100 mg daily.” Ex. 1006, 166–67; Ex. 1002 ¶¶ 256, 257, 268–283.

As explained above, we determine the term “clinically proven effective” to be nonlimiting. *See supra* Section II.B.1. But, even if this claim term were construed as Patent Owner suggests (i.e., clinically proven with data demonstrating a statistically significant difference in the efficacy observed between the regimen at issue compared to the standard of care (PO Resp. 37–38)), we agree with Petitioner that Foley inherently discloses the limitation. Pet. 54. That the COMPASS study results may not yet have been published is of no regard, as “[n]ewly discovered results of known processes directed to the same purpose are not patentable because such results are inherent.” *See Bristol-Myers*, 246 F.3d at 1376. There is no dispute that Foley discloses the claimed dosage amounts (Ex. 1006, 166–67), which read on the material claim limitations requiring 2.5 mg rivaroxaban twice daily and 75–100 mg aspirin daily; that those amounts were not yet identified as “clinically proven effective” does not change those dosage amounts required or otherwise limit the claimed method. *See id.* at 1375. There is no dispute (*see* Pet. 52–58; PO Resp. 52–54) that Foley discloses a method including administering to the appropriate patient group the claimed dosage amounts of rivaroxaban and aspirin at the claimed regimen. Accordingly, we find Foley anticipates claims 1 and 2 of the ’310 Patent.

Patent Owner disagrees with Petitioner’s inherency argument, asserting that a POSA would not have considered the results of a clinical trial to be inherent. PO Resp. 53. Specifically, Patent Owner asserts that “[n]umerous factors can impact whether a dosing regimen will be ‘clinically proven effective,’ including the inclusion and exclusion criteria for the

patients to be studied, and the design and conduct of the study itself.” *Id.* (citing Ex. 2060 ¶¶ 153–157; Ex. 2029, 45:4–13). Stated differently, Patent Owner argues that because the design of clinical trials can vary, the dosing regimen of rivaroxaban and aspirin disclosed in Foley would not necessarily be found to be “clinically proven effective.” Patent Owner further argues that “there is nothing about the COMPASS protocol that makes clinical proof of efficacy ‘necessarily present’ and ‘recognized’ by the POSA.” *Id.* at 54 (citing Ex. 2060 ¶¶ 151–157).

Patent Owner again, however, misses the point. The question is not whether clinical trials may exist where the disclosed dosing regimen may not be clinically proven effective, or whether a POSA would necessarily recognize clinical proof of efficacy from the COMPASS protocol taught by Foley. *Id.* at 53–54. On the contrary, the question is whether “the [prior art] disclosure is sufficient to show that the natural result flowing from the operation as taught would result in the performance of the questioned function.” *King Pharms., Inc. v. Eon Labs, Inc.*, 616 F.3d 1267, 1275 (Fed. Cir. 2010) (citation omitted). If it is, the Federal Circuit has held that it is “well-settled that the disclosure should be regarded as sufficient.” *Id.*

As such, here, we consider whether the disclosure in Foley is sufficient to show that the natural result of the disclosed dosing regimen would be “clinically proven effective” by the COMPASS study disclosed by Foley. In this case, we know that to be true because the claimed process was confirmed by the *same* COMPASS study disclosed by Foley. *See Bristol-Myers*, 246 F.3d at 1376; *see also Abbott Labs. v. Baxter Pharm. Prods., Inc.*, 471 F.3d 1363, 1369 (Fed. Cir. 2006) (finding method claims anticipated where prior art method disclosed the same steps in furtherance of the same purpose, and all that was contributed by the patent claims is the

recognition of a new property of the prior art process). Accordingly, even if the claim term were limiting, we find Foley inherently discloses “clinically proven effective” amounts of rivaroxaban and aspirin.

Having considered the arguments and evidence presented at trial, we find Petitioner has shown by a preponderance of the evidence that claims 1 and 2 are anticipated by Foley.

D. Obviousness over Foley Alone or in Combination with Plosker

Petitioner asserts that claims 1, 2, 5, and 8 are unpatentable as obvious over Foley. Pet. 58–69. Petitioner also asserts that claims 3, 4, 6, and 7 are unpatentable as obvious over Foley and Plosker. *Id.* at 69–71. Patent Owner opposes. PO Resp. 54–56.

Having considered the arguments and evidence presented at trial, we determine Petitioner has established by a preponderance of the evidence that claims 1–8 are unpatentable as obvious over the cited art. We incorporate here our prior discussion and findings regarding Foley.

1. Plosker (Ex. 1007)

Plosker is a journal article entitled “Rivaroxaban: A review of Its Use in Acute Coronary Syndromes,” which was published in 2014. Ex. 1007, 451. Patent Owner does not dispute that Plosker is prior art. *See generally* PO Resp.

Plosker describes the ATLAS trial, including the dosing regimen of 2.5 mg rivaroxaban twice daily, co-administered with 75–100 mg aspirin alone or aspirin plus either clopidogrel or ticlopidine. Ex. 1007, 451, 460. Plosker notes the dosing regimen was approved for use in the EU for patients with a recent ACS. *Id.* at 451.

2. *Analysis*

A patent claim is unpatentable under 35 U.S.C. § 103 if the differences between the claimed invention and the prior art are such that the claimed invention, as a whole, would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. 35 U.S.C. § 103; *see also KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and (4) objective evidence of nonobviousness. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

In analyzing the obviousness of a combination of prior art elements, it can be important to identify a reason that would have prompted one of skill in the art “to combine . . . known elements in the fashion claimed by the patent at issue.” *KSR*, 550 U.S. at 418. A precise teaching directed to the specific subject matter of a challenged claim is not necessary to establish obviousness. *Id.* Rather, “any need or problem known in the field of endeavor at the time of invention and addressed by the patent can provide a reason for combining the elements in the manner claimed.” *Id.* at 420. Accordingly, Petitioner must show that “a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.” *In re Magnum Oil Tools Int'l, Ltd.*, 829 F.3d 1364, 1381 (Fed. Cir. 2016) (quotations and citations omitted).

Petitioner asserts that claims 1, 2, 5, and 8 are unpatentable as obvious over Foley. Pet. 58–69. As explained above, we find Petitioner has shown

by a preponderance of the evidence that Foley anticipates claims 1 and 2. *See supra* Section II.C. Thus, we find Petitioner has also shown by a preponderance of the evidence that Foley renders claims 1 and 2 unpatentable as obvious, as it is well settled that anticipation is the epitome of obviousness.¹³ *See Realtime Data, LLC v. Iancu*, 912 F.3d 1368, 1373 (Fed. Cir. 2019) (“[I]t is well settled that a disclosure that anticipates under § 102 also renders the claim invalid under § 103, for anticipation is the epitome of obviousness.”) (internal quotations and citations omitted).

Petitioner asserts the only difference between claims 1 and 2 and claims 5 and 8 is that the latter recites the “first product comprises 2.5 mg rivaroxaban and 75–100 mg aspirin” (or 100 mg aspirin for claim 8) and the “second product comprises 2.5 mg rivaroxaban.” *Compare* Ex. 1001, claim 1 *with* claim 5. Pet. 66–69. Petitioner therefore asserts that Foley discloses each limitation of claims 5 and 8 except the “first product” and “second product,” for the reasons stated above. *See supra* Section II.C.2 (finding Foley discloses each limitation of claims 1 and 2).

Petitioner notes that Foley does not specify whether the 100 mg aspirin was taken with the first 2.5 mg dose of rivaroxaban, with the second 2.5 mg dose of rivaroxaban, or at a different time than either dose of rivaroxaban. Pet. 67 (citing Ex. 1002 ¶¶ 336–338). Nevertheless, Petitioner asserts that a POSA would have understood that “unless there is a cause for concern in terms of potential drug interactions, concomitant administration of multiple drugs in a dosing regimen is preferred for patient-compliance reasons.” *Id.* (citing Ex. 1002 ¶¶ 337–338).

¹³ We further note that we have considered, but are not persuaded that Patent Owner’s objective evidence of nonobviousness, discussed below, outweighs this strong evidence of obviousness.

Although Patent Owner disagrees with Petitioner’s construction of “first product,” Patent Owner does not argue that Foley does not teach a “first product.”¹⁴ *See* PO Resp. 51–56; PO Sur-reply 19–24. As set forth above, we agree with Petitioner’s construction of “first product” as not requiring, although allowing, a single combination dosage form. *See supra* Section II.B.2. We, therefore, find Petitioner has shown that a POSA considering Foley would have found it obvious to administer aspirin together with one of the rivaroxaban doses. We credit the testimony of Petitioner’s expert, Dr. Zusman, who notes that a POSA would have known that there were no concerns about administering rivaroxaban and aspirin together, and that it is easier for patients to take two drugs at the same time rather than at different times throughout the day. *See* Ex. 1002 ¶¶ 337–338; *see also* Ex. 1009,¹⁵ 981 (noting “there is no clinically relevant interaction between rivaroxaban and aspirin and that the 2 drugs could be administered concomitantly”). We, therefore, find Petitioner has shown that Foley teaches or suggests each limitation of claims 5 and 8 and that a POSA would have had a reason to modify Foley to administer rivaroxaban together with aspirin with a reasonable expectation of success.

In its Response, Patent Owner argues that a POSA would not have had a reasonable expectation of success that the claimed dosing regimen in COMPASS would be “clinically proven effective.” PO Resp. 55 (citing

¹⁴ Although Dr. Weitz appears to argue that Foley does not teach a “first product” (Ex. 2060 ¶ 163), Patent Owner does not make this argument in its papers. *See generally* PO Resp.; PO Sur-reply.

¹⁵ Kubitza et al., *Safety, Tolerability, Pharmacodynamics, and Pharmacokinetics of Rivaroxaban—an Oral, Direct Factor Xa Inhibitor—Are Not Affected by Aspirin*, 46 J. CLINICAL PHARMACOLOGY 981–90 (2006) (Ex. 1009).

Ex. 2060 ¶¶ 158–165). We are not persuaded, however, because, as explained above, the term “clinically proven effective” is inherent in the prior art teaching, but, in any event, adds no patentable weight to the material amounts recited in the claims. *See supra* Section II.B.1.

We find the Federal Circuit’s decision in *Persion Pharmaceuticals LLC v. Alvogen Malta Operations Ltd.*, 945 F.3d 1184 (Fed. Cir. 2019) to be instructive. In that case, the claims recited, *inter alia*, a method of treating pain in a patient having mild or moderate hepatic impairment, comprising administering an oral dosage unit comprising an extended release formulation of hydrocodone bitartrate, wherein the dosage unit provides a specific pharmacokinetic release profile of hydrocodone. *Id.* at 1187. The district court determined the claims were invalid as obvious because a POSA would have been motivated to administer the extended-release hydrocodone formulation disclosed in an asserted Devane reference to patients with mild or moderate hepatic impairment, as taught by the other asserted prior art references. *Id.* at 1188–89. The district court held that “the pharmacokinetic limitations . . . are ‘inherent in any obviousness combination that contains the Devane formulation’ because the recited pharmacokinetic parameters were ‘necessarily present’ in the Zohydro ER [i.e., extended release hydrocodone] formulation described in both Devane and the asserted patents.” *Id.* at 1189.

On appeal, the Federal Circuit affirmed, noting that “[i]nherency may supply a missing claim limitation in an obviousness analysis.” *Id.* at 1190. The court also noted that “[i]t is long settled that in the context of obviousness, the ‘mere recitation of a newly discovered function or property, inherently possessed by things in the prior art, does not distinguish a claim drawn to those things from the prior art.’” *Id.* (quoting *In re Oelrich*, 666

F.2d 578, 581 (CCPA 1981). The patentee argued that the district court erred in applying inherency to its obviousness analysis because the “natural result flowing from the operation as taught” by Devane cannot be the claimed pharmacokinetic values for hepatically impaired patients because Devane does not teach administering its hydrocodone to hepatically impaired patients. *Id.* (quoting *Oelrich*, 666 F.2d at 581). The Federal Circuit rejected this argument, because “[t]here was also no dispute that the Devane formulation, which was identical to the Zohydro ER formulation described in the patents in suit, necessarily exhibited the claimed parameters under these conditions.” *Id.* Accordingly, the Federal Circuit held the district court did not err in finding the pharmacokinetic limitations of the asserted claims to be “inherent and added no patentable weight to the pharmacokinetic claims.” *Id.*

The case for inherency is even stronger here than in *Persion*, because there is no need to combine multiple references to determine “the natural result flowing from the operation as taught” by Foley. Rather, Foley discloses a method including the exact dosing regimen, drugs, and patients described and claimed in the ’310 Patent. Thus, like the Zohydro ER formulation of Devane and the patents-in-suit in *Persion*, we find that the “natural result flowing from the operation as taught” in Foley is necessarily a finding of “clinically proven effective” amounts of rivaroxaban and aspirin. *Id.* at 1191.

Claims 3 and 7 depend from independent claims 1 and 5, respectively, and recite 81 mg aspirin, whereas claims 4 and 6 depend from independent claims 1 and 5, respectively, and recite 75 mg aspirin daily. Petitioner asserts that Foley teaches administering 100 mg aspirin, and Plosker teaches administering 75–100 mg aspirin to patients with ACS. Pet. 70 (citing

Ex. 1006, 166–67; Ex. 1007, 460). Petitioner asserts that from these teachings, a POSA “would have had good reason to administer to a human patient with CAD or PAD a dosage of aspirin that falls within the 75–100 mg range of aspirin taught by Plosker.” *Id.* at 70–71 (citing Ex. 1002 ¶¶ 353–355, 359–361). As noted by Petitioner, Dr. Zusman also explains that “no clinical difference exists between administering 100 mg daily aspirin, as expressly taught by Foley, and 81 or 75 mg daily aspirin” and that these dosages “just represent the strength of commercially available dosage units in different areas of the world.” *Id.* at 71 (citing Ex. 1002 ¶¶ 351–362; Ex. 1013, 2656).

Patent Owner does not challenge Petitioner’s assertions with respect to these additional limitations of the claims. Rather, Patent Owner relies on its arguments regarding claims 1 and 5 for why claims 3, 4, 6, and 7 would not have been obvious. *See* PO Resp. 56.¹⁶

We are persuaded that a POSA would have had a reason to use 75 mg and 81 mg aspirin daily instead of 100 mg aspirin taught by Foley with a reasonable expectation of success, because those dosages simply represent the dosage amounts of aspirin available throughout the world (i.e., aspirin is available in Germany in 75 and 100 mg dosages, and 81 mg is more common in the United States)—hence, obvious variant dosages. *See* Ex. 1013, 2656; *see also* Ex. 1002 ¶ 355; Ex. 1001, 9:21–22 (stating the

¹⁶ Patent Owner argues “for all the reasons that claims 1 and 5 are not obvious over . . . Foley (or Foley in combination with Plosker . . .), the POSA would not regard claims 3, 4, 7, or 8 to be obvious over the combination of Foley and Plosker.” PO Resp. 56. We understand Patent Owner to mean “claims 3, 4, 6, *or* 7,” as recited in the title of Section IX of Patent Owner’s Response (“CLAIMS 3, 4, 6, and 7 ARE NOT OBVIOUS OVER FOLEY AND PLOSKER (GROUND 5)”). *See id.*

individual active ingredients “are commercially available”). Moreover, those dosages are consistent with the dosage range taught by Plosker. *See* Ex. 1007, 460.

Accordingly, having considered the arguments and evidence presented at trial, we find Foley alone or the combination of Foley and Plosker teach or suggest each limitation of claims 1–8 and a POSA would have had a reasonable expectation of success in reaching the claimed invention.

In reaching our conclusion on obviousness, we considered, and we now turn to discuss, Patent Owner’s objective evidence of nonobviousness.

3. *Objective Evidence of Nonobviousness*

“Objective indicia of nonobviousness can serve as an important check against hindsight bias and ‘must always when present be considered.’”

Merck & Cie v. Gnosis S.P.A., 808 F.3d 829, 837 (Fed. Cir. 2015) (citation omitted). Patent Owner argues unexpected results support the patentability of the claims. PO Resp. 57–60; PO Sur-reply 24–25. “To be particularly probative, evidence of unexpected results must establish that there is a difference between the results obtained and those of the closest prior art, and that the difference would not have been expected by one of ordinary skill in the art at the time of the invention.” *Bristol-Myers Squibb Co. v. Teva Pharms USA, Inc.*, 752 F.3d 967, 977 (Fed. Cir. 2014).

Patent Owner argues that a POSA would regard the COMPASS regimen disclosed, for example, on the ClinicalTrials.gov website, as the closest prior art. PO Resp. 57 (citing Ex. 2001, BAYX310-0001400-003; Ex. 2034, 1–4; Ex. 2060 ¶¶ 158–174). Patent Owner asserts, however, that a POSA would not have had a reasonable expectation that the use of 2.5 mg rivaroxaban twice daily plus low-dose aspirin once daily would be clinically proven effective in reducing the risk of myocardial infarction, stroke, or

cardiovascular death in a human patient with CAD and/or PAD, as required by the claims. *Id.* Patent Owner further asserts that it was unexpected and surprising that the COMPASS trial demonstrated clinical proof of efficacy, but that “the method was so efficacious that the trial was stopped early because the results were so positive.” *Id.* (citing Ex. 2060 ¶¶ 168–174). Patent Owner notes that as of the priority date, no other antiplatelet and anticoagulation combination was established for use in patients with CAD and/or PAD and there was an unmet need for more effective therapy in those patients, which demonstrates a difference in kind rather than degree. *Id.* at 58–59 (citing Ex. 2060 ¶ 171).

As an initial matter, Petitioner notes that Patent Owner “blurs its unexpected results argument with its reasonable expectation of success argument with the circular presumption that the results were unexpected because they were not reasonably expected.” Pet. Reply 18 n.4. Moreover, Petitioner disagrees that early termination of the COMPASS trial constitutes evidence of nonobviousness. Pet. Reply 18–19. Petitioner argues that “the *scientific* result was not unexpected: COMPASS merely confirmed the expected scientific result sooner than its study design required.” *Id.* (citing Ex. 1037 ¶ 71). Moreover, Petitioner asserts that ATLAS had already shown the claimed method was safe and, as Dr. Zusman explains, “cardiac trials often stop early as the study design permits when prespecified rules are satisfied.” *Id.* at 20 (citing Ex. 1038 ¶¶ 76–81).

Having fully considered Patent Owner’s objective evidence of unexpected results, we find Petitioner has the better position. As an initial matter, we note that Patent Owner’s arguments and evidence of unexpected results rely solely on the term “clinically proven effective,” as recited in the claims. *See, e.g.*, PO Resp. 57 (arguing “the POSA would not have had a

reasonable expectation that [the COMPASS regimen] would be clinically proven effective”); PO Sur-reply 24 (arguing “the fact that the combination of aspirin and rivaroxaban in COMPASS actually showed clinical proof of efficacy was unexpected”). As explained above, however, we determine “clinically proven effective” to be nonlimiting, and, even if considered to be limiting, it was inherent in view of the prior art. *See supra* Sections II.B.1, II.C.2, II.D.2.

That said, even if we considered the term “clinically proven effective” to carry some patentable weight, our Decision does not change. Although we agree that ending the COMPASS trial early was unexpected, we are not persuaded that that fact alone was an unexpected result that is probative of nonobviousness. As the Federal Circuit has held, “[t]o be particularly probative, evidence of unexpected results must establish that there is a difference between the results obtained and those of the closest prior art, and that the difference would not have been expected by one of ordinary skill in the art at the time of the invention.” *UCB, Inc. v. Actavis Labs. UT, Inc.*, 65 F4th 679, 693 (Fed. Cir. 2023). Moreover, “[a] difference of degree is not as persuasive as a difference in kind—i.e., if the range produces a new property dissimilar to the known property, rather than producing a predictable result but to an unexpected extent.” *Id.* (internal quotations omitted).

Here, we find terminating the COMPASS study early based on the “consistent difference in the primary efficacy outcome” amounts to a difference in degree rather than a difference in kind. *See* Ex. 1001, 15:41–47. That is, as Dr. Zusman explains, “[a] clinical trial may terminate early due to benefit if ‘[t]he study hypothesis is unexpectedly proven early within predesignated criteria.’” Ex. 1037 ¶ 77. Thus, we agree with Patent Owner that the results’ rapidity may have been unexpected, or else the

“‘prespecified’ criteria would have been the endpoints in the first instance rather than [the] criteria for early termination.” PO Sur-reply 25. Despite those unexpectedly rapid results, however, we are not persuaded that ending the trial early necessarily amounts to a difference in kind. In other words, the results do not demonstrate “a new property dissimilar to the known property.” *UCB*, 65 F.4th at 693. Instead, we find the study results to be more akin to “producing a predictable result but to an unexpected extent.” *Id.*

Moreover, when considered together with the other obviousness evidence presented at trial, we conclude that Patent Owner’s evidence of objective indicia does not outweigh the strong evidence of obviousness discussed above. *Asyst Techs., Inc. v. Emtrak, Inc.*, 544 F.3d 1310, 1316 (Fed. Cir. 2008) (“[A]s we have often held, evidence of secondary considerations does not always overcome a strong prima facie showing of obviousness.”); *see also Leapfrog Enters., Inc. v. Fisher-Price, Inc.*, 485 F.3d 1157, 1162 (holding substantial evidence of commercial success, praise, and long-felt did not outweigh the strength of the prima facie obviousness showing).

4. *Conclusion on Obviousness of Claims 1–8*

Having considered the arguments and evidence presented at trial, we find Petitioner has shown by a preponderance of the evidence that claims 1, 2, 5, and 8 are unpatentable as obvious over Foley and claims 3, 4, 6, and 7 are unpatentable as obvious over Foley and Plosker.

E. *Remaining Grounds*

Petitioner asserts that claims 1–4 are anticipated by EMA. Pet. 21–33. Petitioner also asserts that claims 1–8 are unpatentable as obvious over EMA. Pet. 33–51. Because we have determined claims 1–8 of the ’310

Patent are unpatentable as anticipated and/or obvious over Foley alone or in combination with Plosker and therefore dispositive as to all challenged claims, we need not reach the grounds over EMA for purposes of our Decision. *See SAS Inst. Inc. v. Iancu*, 138 S. Ct. 1348, 1359 (2018) (holding that a petitioner “is entitled to a final written decision addressing all of the claims it has challenged”); *Bos. Sci. Scimed, Inc. v. Cook Grp. Inc.*, 809 F. App’x 984, 990 (Fed. Cir. 2020) (non-precedential) (recognizing that the “Board need not address issues that are not necessary to the resolution of the proceeding” and, thus, agreeing that the Board has “discretion to decline to decide additional instituted grounds once the petitioner has prevailed on all its challenged claims”).

III. PETITIONER’S MOTION TO EXCLUDE

The party moving to exclude evidence bears the burden of proving that it is entitled to the relief requested—namely, that the material sought to be excluded is inadmissible under the Federal Rules of Evidence (“FRE”). *See* 37 C.F.R. §§ 42.20(c), 42.62(a).

Petitioner filed a Motion to Exclude Exhibits 2030, 2043–2047, 2060, and 1030 (125:9–130:25). Paper 59, 1. Patent Owner opposes Petitioner’s Motion. Paper 61. Because we determine claims 1–8 of the ’310 Patent are unpatentable even considering the objected-to evidence, we dismiss Petitioner’s Motion to Exclude as moot.

IV. CONCLUSION¹⁷

For the foregoing reasons, we determine that Petitioner has established by a preponderance of the evidence that claims 1–8 of the '310 Patent are unpatentable.

In summary:

Claim(s)	35 U.S.C. §	Reference(s)/Basis	Claim(s) Shown Unpatentable	Claim(s) Not shown Unpatentable
1, 2	102	Foley	1, 2	
1, 2, 5, 8	103	Foley	1, 2, 5, 8	
3, 4, 6, 7	103	Foley and Plosker	3, 4, 6, 7	
1–4	102	EMA ¹⁸		
1–8	103	EMA ¹⁹		
Overall Outcome			1–8	

V. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claims 1–8 of U.S. Patent No. 10,828,310 B2 are held unpatentable;

¹⁷ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. See 37 C.F.R. § 42.8(a)(3), (b)(2).

¹⁸ As explained above, we need not reach the grounds involving EMA because the grounds over Foley are dispositive of all challenged claims.

¹⁹ As explained above, we need not reach the grounds involving EMA because the grounds over Foley are dispositive of all challenged claims.

FURTHER ORDERED that Petitioner's Motion to Exclude is
dismissed as moot; and

FURTHER ORDERED that, because this is a Final Written Decision,
the parties to the proceeding seeking judicial review of the decision must
comply with the notice and service requirement of 37 C.F.R. § 90.2.

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Patent 10,828,310 B2

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